

Vol 4 Chapter 1 — Newton’s G from Bergman Curvature

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Chapter 1 — Newton’s G from Bergman Curvature

Why this chapter matters

Newton’s gravitational constant G is the weakest of the four fundamental couplings. The dimensionless gravitational fine-structure analog $\alpha_G \approx 10^{-39}$ is thirty-seven orders of magnitude below the electromagnetic $\alpha \approx 10^{-2}$. This thirty-seven-order hierarchy between gravitational and electromagnetic coupling strengths is what physicists call the hierarchy problem, and it is among the deepest unexplained features of fundamental physics. Standard physics has no derivation of G ; it is taken as an experimental input, measured by Cavendish torsion balances and refined over two centuries.

BST derives Newton’s G in BST-primary form, to 0.07% precision, with no fitted parameters:

$$G = \hbar c \cdot (6\pi^5)^2 \cdot \alpha^{24} / m_e^2.$$

Every factor on the right is substrate-derived. The $(6\pi^5)^2$ is the square of the proton-to-electron mass ratio of Volume 2 Chapter 6. The $\alpha = 1/N_{\max} = 1/137$ is the substrate’s fine-structure cutoff of Chapter 10 in Volume 1. The exponent $24 = 4 \cdot C_2$ — four complete Bergman round trips around the substrate’s $C_2 = 6$ Casimir modes — is forced by the substrate’s root-system structure. The electron mass m_e sets the unit scale. There is no tunable freedom; the gravitational coupling is computed from BST primaries.

The match to measurement is $6.66961 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$ versus the measured $6.67430 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$ — a deviation of 0.07%, well within the experimental precision of present-day Cavendish-type torsion-balance measurements. The thirty-seven-order gravity-versus-electromagnetism hierarchy, which standard physics treats as a mysterious fine-tuning, in BST falls out structurally: gravity needs *four* Bergman round trips per coupling vertex, electromagnetism needs only *two*, so gravity is suppressed relative to electromagnetism by $\alpha^{4 \cdot C_2 - 2 \cdot C_2} = \alpha^{12}$ — which combined with the proton-mass factors gives the observed hierarchy.

This chapter sets out the derivation, the substrate-mechanism Bergman-round-trip framework, three independent structural confirmations of the exponent 24, and the substrate-side reading of gravity as the residual curvature of the substrate’s commitment computation.

1.1 The T1296 substrate-derivation

Lyra T1296 (April 2026, “Gap #1 closed”), with computational verification in Elie’s Toy 541 and the heat-kernel cross-check in Toy 639, establishes the substrate-mechanical derivation of Newton’s G :

$$G = \hbar c \cdot (6\pi^5)^2 \cdot \alpha^{24} / m_e^2.$$

The factors are each substrate-mechanical:

- $(6\pi^5)^2 = (m_p/m_e)^2$. The proton-to-electron mass ratio from Volume 2 Chapter 6 enters squared because gravity couples to two masses (source and probe). At leading order, $(6\pi^5)^2 \approx 1836.118^2 \approx 3.37 \times 10^6$.

- α^{24} . The substrate’s fine-structure coupling $\alpha = 1/N_{\max} = 1/137$, raised to the twenty-fourth power. The exponent is what makes gravity weak: $\alpha^{24} = (1/137)^{24} \approx 10^{-51.4}$, an extremely small suppression that pulls G down from any naturally-scaled value to its observed extraordinarily small magnitude.
- $1/m_e^2$. Dimensional. Gravity scales as inverse mass squared in natural units; the electron mass sets the substrate’s unit-scale floor.

The combination gives G in natural units, which converts to SI via the standard factors. Toy 541 (the framework’s “five integers to everything” computational reproduction, 16/16 sub-checks passing) confirms the numerical match at 0.07%.

The match precision is, for a gravitational measurement, excellent. Newton’s G has been measured to about parts-per-thousand precision by modern Cavendish-type experiments; substantial systematic discrepancies between independent measurements exist at the parts-per-ten-thousand level. The substrate prediction sits comfortably within current experimental precision.

1.2 The Bergman round-trip mechanism

The exponent $24 = 4 \cdot C_2$ has a substrate-mechanism reading that is the chapter’s structural content.

The substrate Hilbert space $H^2(D_{IV}^5)$ admits a Bergman round-trip framework, established in the BST research note `BST_AlphaSquared_LayerProof.md` and connecting back to the substrate’s S^1 phase structure on K -types. The key idea: each “Bergman round trip” corresponds to one complete phase circuit on the substrate’s holomorphic structure, with the forward and return half-cycles each contributing a factor of $\alpha^{1/2}$ to the amplitude (via the Szegő-kernel factorization on D_{IV}^5 ’s boundary). The round-trip *probability* is therefore α , the round-trip *amplitude* is $\alpha^{1/2}$, and the suppression factor for n round trips at k Casimir modes each is $\alpha^{n \cdot k}$.

For electromagnetism, the substrate coupling requires *two* Bergman round trips at the EM Casimir level, giving $\alpha^2 \approx 5.3 \times 10^{-5}$ as the EM suppression per vertex. This matches QED’s familiar α^2 scaling of higher-order processes.

For gravity, the substrate coupling requires *four* Bergman round trips at $C_2 = 6$ Casimir modes each, giving $\alpha^{4 \cdot 6} = \alpha^{24}$ as the gravitational suppression per vertex. The factor of four comes from the substrate’s $\mathfrak{so}(5, 2)$ structure: gravity must close four independent loops in the substrate’s representation cohomology — one for each of the four positive roots of the B_2 root system that we used in Chapter 5 of Volume 1 to define the Casimir eigenvalue formula.

So the structural reason gravity is much weaker than electromagnetism is *not* a fine-tuning of G versus α . It is the substrate’s requirement that gravitational coupling close four times as many independent loops as electromagnetic coupling. The hierarchy α^{22} between the two — twelve additional powers of α — is what produces the observed thirty-seven orders of magnitude between α_G and α .

1.3 Three independent confirmations of $24 = 4 \cdot C_2$

The exponent 24 in the substrate’s derivation of G is not arbitrary. It is forced by three independent structural arguments, each of which gives the same answer.

Confirmation 1 — Heat kernel at level $k = 16$. The heat-kernel expansion on D_{IV}^5 has a speaking-pair structure with period $n_C = 5$. At the third speaking pair, $k = 16$, the substrate’s heat-kernel ratio evaluates to

$$\text{ratio}(k = 16) = -24 = -\dim SU(5) = -(n_C^2 - 1).$$

This is Toy 639’s confirmed computational result (May 2025). The number 24 shows up as $n_C^2 - 1$ — the dimension of the group $SU(n_C) = SU(5)$ — at the heat-kernel level where the speaking-pair structure makes the relevant coefficient extractable.

Confirmation 2 — Uniqueness identity $n_C^2 - 1 = (n_C - 1)!$. The arithmetic identity $n_C^2 - 1 = (n_C - 1)!$ holds *uniquely* at $n_C = 5$: $5^2 - 1 = 24 = 4! = (5 - 1)!$. For $n_C = 4$: $15 \neq 6$. For $n_C = 6$: $35 \neq 120$. For $n_C = 7$: $48 \neq 720$.

The identity locks the substrate’s gravitational exponent to its specific complex dimension; if n_C were anything other than 5, the identity would fail and the exponent would not be 24.

Confirmation 3 — Casimir decomposition $24 = 4 \cdot C_2$. The exponent 24 can be read as four complete cycles through the substrate’s $C_2 = 6$ eigenvalues — what we called the four Bergman round trips in §1.2 — at heat-kernel levels $k = 6, 12, 18, 24$. The four-fold periodicity in Casimir-eigenvalue space is structural; it matches the four positive roots of the substrate’s B_2 root system.

Three independent BST primary readings converge on 24. The substrate-derivation is over-determined; the value of the exponent is not a fitting parameter. This is the same pattern of substrate over-determinism that we identified throughout Volume 0 and Volume 1: distinct structural constraints converging on the same answer.

1.4 Gravity as residual substrate curvature

Lyra T2106 (“Gravity as Eigentone,” April 2026) provides a deeper substrate-mechanism reading: gravity is what remains of the substrate’s curvature *after* all channel commitments have run their per-tick computations to completion.

The substrate’s four-phase commitment cycle (Volume 0 Chapter 3) processes substrate state through Zones 1 to 4 on each Koons tick. In the strong, electromagnetic, and weak sectors, the substrate’s per-tick computation closes the relevant K-type commitments — quarks confine, electromagnetic amplitudes propagate, weak decays proceed — within a definite number of Bergman round trips at the level appropriate to each sector. Each sector’s coupling reflects its commitment depth: two round trips for EM, three for weak, three for strong (with color confinement at $C_2 = 6$ modes each).

Gravity sits at the *outside* of this commitment structure. After the strong, electromagnetic, and weak sectors have completed their per-tick commitments, the substrate’s residual curvature — the un-committed portion of the substrate’s Bergman geometry — is what couples masses gravitationally. The four Bergman round trips at $C_2 = 6$ modes are the residual round trips that gravity inherits: they are what is *left over* after the other three sectors have run.

This is the substrate-mechanism reading of why gravity does not unify with the other forces at any finite energy scale. The other forces commit; gravity is the residual. The substrate produces gravitational coupling not from a unified gauge sector but from the *complement* of the gauge sector’s commitment depth. The substrate’s no-GUT prediction (Volume 1 Chapter 8’s Five-Absence set) and the substrate’s gravity-as-residual reading are the same structural fact viewed from two sides.

A consequence is that BST’s four-dimensional general relativity emerges from the substrate’s six-dimensional internal Bergman computation (the integrating-out of the substrate’s K-type internal degrees of freedom gives the residual four-dimensional curvature that becomes Einstein’s tensor). T2106’s full statement is that four-dimensional Newton’s G is exactly the integrated residual curvature of the substrate, computed via Bergman integration over the substrate’s internal six-dimensional structure. Volume 4 Chapter 2 develops this in detail.

1.5 What the chapter buys

The substrate framework, by deriving Newton’s G to 0.07% from BST primaries with the hierarchy problem resolved structurally, removes one of the deepest unexplained features of standard physics. There is no fine-tuning of G against α . There is no unexplained gravitational weakness. The thirty-seven orders of magnitude between gravity and electromagnetism are the substrate’s required commitment-depth difference: four Bergman round trips for gravity, two for electromagnetism, giving α^{22} suppression that produces the observed hierarchy.

The match at 0.07% is well within current experimental precision. The substrate framework’s prediction is, in this case, more precise than the disagreements between independent Cavendish-type measurements of G — the framework is, structurally, at the experimental frontier.

1.6 What comes next

Chapter 2 — Gravity as Cumulative Eigentone Effect — develops the substrate-mechanism reading of T2106 in detail, showing how the residual substrate curvature integrates to produce the Einstein field equations of general relativity.

Chapter 3 — BST-SR / BST-GR Boundary — examines where in the substrate’s structure the transition from special-relativistic to general-relativistic behavior occurs.

Chapter 4 — The Cosmological Constant from Substrate — gives the substrate-derivation of $\Lambda \approx 10^{-121}$ from the Zone-2 vacuum that we sketched in Volume 0 Chapter 5, with the substrate-mechanism unification of Λ and the laboratory Casimir effect.

Chapters 5 through 12 develop substrate-derivations of the Hubble constant (with the famous Hubble tension resolved via four routes), CMB structure (the scalar spectral index $n_s = 1 - n_C/N_{\max}$ at 0.3σ precision), inflation parameters, big-bang nucleosynthesis (including the Li-7 problem), the cosmological cycle hypothesis and the Interstasis period between Big Bang cycles, dark matter and dark energy (with $\Omega_\Lambda = 13/19$ at 0.07σ and dark-matter-to-baryon ratio $16/3$ at 0.58%), gravitational waves (with NANOGrav predictions), and the SP-27 observational reanalysis program.

The reader who has accepted the substrate-derivation of G in this chapter — the cleanest of the volume’s GR-side derivations — will find the rest of Volume 4 a series of similar substrate-mechanical derivations applied to specific cosmological observables. The substrate produces the entire cosmology side of physics from the same five integers that produce the particle physics side.

Where to look this up: The primary derivation is Lyra T1296 (“Gap #1 closed,” April 2026, with the supporting writeup at `notes/BST_EinsteinEquations_FromCommitment.md`). The Bergman round-trip framework is documented at `notes/BST_AlphaSquared_LayerProof.md`. The three confirmations of the exponent 24 are: Toy 639 (heat kernel at $k = 16$ with ratio $= -24 = -\dim SU(5)$, May 2025), the uniqueness identity $n_C^2 - 1 = (n_C - 1)!$ at $n_C = 5$ (elementary arithmetic check), and the Casimir decomposition $24 = 4 \cdot C_2$ (substrate root-system structure). The gravity-as-eigentone reading is Lyra T2106 (April 2026). The computational verification is Toy 541 (16/16 sub-checks passing). For the standard-physics hierarchy problem and the cosmological-constant problem, the canonical reviews are Weinberg 1989, “The cosmological constant problem,” *Reviews of Modern Physics* 61:1; and Arkani-Hamed et al. 1998 on the hierarchy problem from the modern-extra-dimensions perspective. For the experimental status of Newton’s G measurements with the systematic discrepancies, see the CODATA 2018 evaluation and the Particle Data Group’s *Review of Particle Physics* annual update.